

KARTHICK RAJA S

Embedded Software Intern | Software Development Enthusiast

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PROFESSIONAL SUMMARY

Motivated Electronics Engineering graduate currently working as an Embedded Software Intern at Mistral Solutions, with hands-on experience in mmWave radar systems and Python-based data visualization. Proficient in C, Python, and MySQL, with basic exposure to full-stack development using React and Node.js. Strong interest in data analysis and intelligent systems, eager to learn and apply data-driven solutions to real-world problems.

SKILLS

Programming Languages: C, Python, Kotlin (Basic)

Web Technologies: HTML, CSS, JavaScript (Learning), React (Learning), Node.js (Learning)

Databases: MySQL

Tools & Platforms: VS Code, IntelliJ IDEA, Android Studio, Eclipse, Jupyter Notebook, Power BI

Design & Others: Figma, Canva, DaVinci Resolve, Qt Designer (Python GUI Development)

EXPERIENCE

Embedded Software Intern

Mistral Solutions Pvt Ltd, Kovilpatti, Tamil Nadu

March 2026 – Present

- Worked on mmWave radar systems (IWR1843AOP, AWR6843AOP) and developed a Python-based visualizer to replicate TI's mmWave demo tool.
- Built a data visualization interface to interpret radar output in real-time.
- Developed a full-stack web application using React, Node.js, and MySQL as part of internal project work.

PROJECTS

Edge Intelligent Air Quality Monitoring with Purification System – Tech: Embedded C, Python

- Designed and developed a system to monitor air quality and trigger purification based on environmental conditions.
- Processed real-time sensor data for air quality analysis.

Crowd Monitoring and Density Detection System – Tech: Python, OpenCV

- Built a crowd density monitoring system using computer vision techniques.
- Processed video/image data to estimate crowd levels in real-time.

Grapevine Disease Detection Model – Tech: Python, ML

- Developed an image classification model to detect grape leaf diseases.
- Trained and evaluated the model using a Kaggle dataset.

EDUCATION

B.E. Electronics and Communication Engineering

Jansons Institute of Technology, Coimbatore | 2022 – 2026